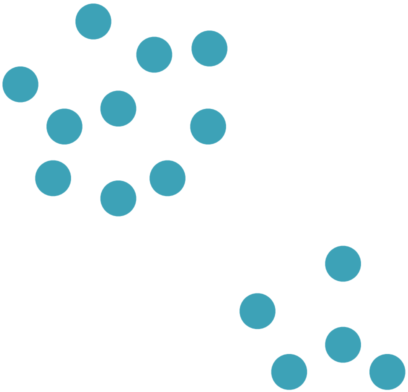
need derivatives w.r.t support points $\mathbf{x}_i$

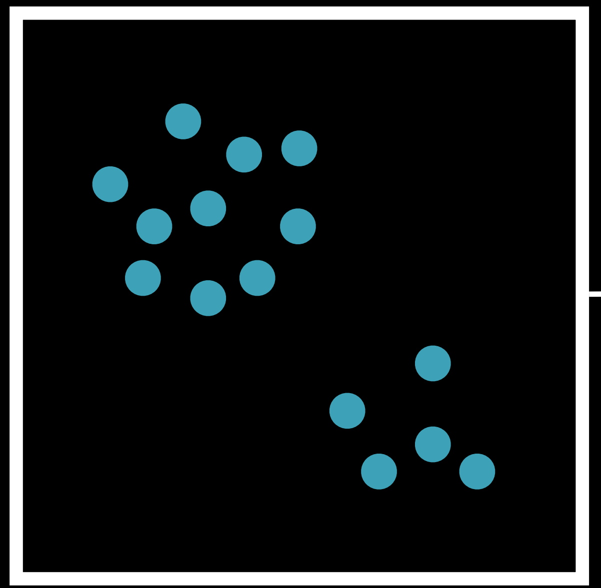


Jacobian of $\mathbf{X}(\theta)$ wrt θ

Gradients wrt all points $1, \ldots, n$ stacked into a $nd \times 1$ vector

Derivatives w.r.t. Cost and Support

Lagrangian (=particles)



$$\frac{\partial \mathrm{OT}}{\partial \mathbf{x}_i}, \frac{\partial \mathrm{OT}}{\partial \mathbf{C}}$$

- For Lagrangian discretization, **need derivatives w.r.t support points $\mathbf{x}_i$**
- $\mathscr{L}(\theta)$ depends on $\mathbf{x}_i$'s via cost matrix $\mathbf{C}$ $\Rightarrow$ need derivative wrt $\mathbf{C}$ first
- For fixed $\mathbf{a}, \mathbf{b}$ and $\varepsilon > 0$, the map $\mathbf{C} \mapsto \mathrm{OT}^{\varepsilon}_{\mathbf{C}}(\mathbf{a}, \mathbf{b})$ is concave + smooth, and

$$\nabla \mathrm{OT}^{\varepsilon}_{\mathbf{C}}(\mathbf{a}, \mathbf{b}) = \pi^{\star}_{\varepsilon}$$

where $\pi^{\star}_{\varepsilon}$ is the unique optimal solution of EOT

- Using the chain rule, we can get derivatives with respect to support points $\mathbf{x}_i$:

$$\nabla_{\mathbf{x}_i} \mathrm{OT}^{\varepsilon}_{c}(\mathbf{a}, \mathbf{b}) = \sum_{j=1}^{m} \pi^{*}_{i,j} \nabla_x c(\mathbf{x}_i, \mathbf{y}_j)$$

- Using chain rule again:

$$\nabla \mathscr{L}(\theta) = [D_\theta\, \mathbf{X}(\theta)]^\top \left(\nabla_{\mathbf{X}} \mathrm{OT}^{\varepsilon}_{c}(\mathbf{a}, \mathbf{b}) \right)$$

Jacobian of $\mathbf{X}(\theta)$ wrt θ

Gradients wrt all points $1, \ldots, n$ stacked into a $nd \times 1$ vector

- Can be used to minimize $\mathscr{L}$ via gradient descent

Pop-Quiz

- Derivatives wrt support points: $\nabla_{\mathbf{x}_i} \mathrm{OT}_c^{\varepsilon}(\mathbf{a}, \mathbf{b}) = \sum_{j=1}^{m} \pi_{i,j}^{*} \nabla_x c(\mathbf{x}_i, \mathbf{y}_j)$

- Show that for $c(x, y) = \|x - y\|^2$, $\nabla_{\mathbf{x}_i} \mathrm{OT}_c^{\varepsilon}(\mathbf{a}, \mathbf{b}) \propto \mathrm{Id} - T_{\mathrm{bary}}$